

THE QUESTION

How do we classify by modelling how each class generates its features, with few enough numbers to estimate?

THE INTUITION

Bayes' rule turns $P(x | y)$ and $P(y)$ into $P(y | x)$. A full $P(x | y)$ needs a number per feature combination, so naive Bayes treats the features as independent given the class: one small table, or one Gaussian, per feature.

KEY TAKEAWAY

$\hat{y} = \arg \max_y P(y) \prod_j P(x_j \mid y)$,
fitted by counting with 1
added and computed as a
sum of logs. A Bayesian
network factorises a joint as
 $\prod_i P(x_i \mid \text{parents}(x_i))$.

labelled data $\rightarrow$ count $P(y), P(x_j | y)$ $\rightarrow$ add 1 to counts $\rightarrow$ $\log P(y) + \sum_j \log P(x_j | y)$ $\rightarrow$ largest score wins

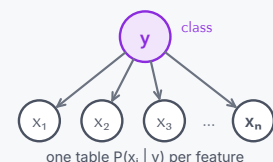
1 Bayes' rule turns a test result around



$P(y | x) = P(x | y)P(y)/P(x)$ with $P(x) = \sum_c P(x | c)P(c)$. The lecture's test is 99% right for the sick and for the healthy, and 1 person in 10,000 is sick. False positives then outnumber the sick 101 to 1, and a positive gives only

$$P(D \mid +) = \frac{0.99 \cdot 10^{-4}}{0.99 \cdot 10^{-4} + 0.01 \cdot 0.9999} = \frac{1}{102} \approx 0.0098.$$

2 Features are independent given the class



Model how each class produces x , then invert. $P(x)$ is common to all classes, so

$$\hat{y} = \arg \max_y P(y) \prod_{i=1}^n P(x_i | y).$$

Naive: given y , one word says nothing about another, $P(x_2 | x_1, y) = P(x_2 | y)$. Rarely true, yet the class ranking is often right, and n binary words need $2n + 1$ numbers, not $2^{n+1} - 1$.

3 Fitting is counting

spam (y = 1)					ham (y = 0)					$P(\mathbf{w}_i = 1 \mid y)$	spam	ham	
w_1	0	0	1	1	0	0	1	1	1	0	w_1	2/5	3/5
w_2	0	0	0	1	1	0	0	1	1	1	w_2	2/5	3/5
w_3	0	0	1	1	1	0	0	1	0	1	w_3	3/5	2/5
#	1	3	5	7	10	2	4	6	8	9	prior $P(y)$	1/2	1/2

Maximum likelihood gives frequencies within each class,

$$P(x_j = 1 \mid y = c) = \frac{\text{count}(x_j = 1, y = c)}{\text{count}(y = c)},$$

and $P(y = c)$ is the class's share of the rows: one pass over the data. The slides' $P(w_1 = 0 \mid \text{ham}) = 3/5$ and $P(w_2 = 1 \mid \text{spam}) = 1/5$ are both $2/5$ in their own table.

4 A new email multiplies one factor per word

test email	$w_1 = 0$	$w_2 = 0$	$w_3 = 1$			divide by the sum
				prior		
spam	$3/5$	$\times$	$3/5$	$\times$	$3/5 \times 1/2$	$= 0.108$
						0.771
ham	$2/5$	$\times$	$2/5$	$\times$	$2/5 \times 1/2$	$= 0.032$
						odds $0.108 : 0.032$
						$= 3.375 : 1$ spam

For the email (0, 0, 1) an absent word contributes $P(w_j = 0 \mid y) = 1 - P(w_j = 1 \mid y)$. Spam scores $(3/5)^3 \cdot 1/2 = 0.108$, ham $(2/5)^3 \cdot 1/2 = 0.032$, so $P(\text{spam} \mid \mathbf{x}) = 0.771$. The slides reach spam from 0.144 and 0.072 (ratio 2); their ham score also uses $P(w_3 = 0 \mid \text{ham})$ for a present w_3 .

5 One zero count vetoes all other evidence

a 4th word w_4 (illustrative):
seen in 0 of 5 spam, 1 of 5 ham



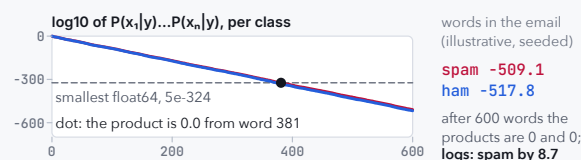
A word never seen with a class gets probability 0, so the product is 0 whatever the other words say.

Laplace smoothing adds $\alpha = 1$ to every count:

$$P(x_j = 1 \mid y = c) = \frac{\text{count}(x_j = 1, y = c) + \alpha}{\text{count}(y = c) + 2\alpha},$$

2 being the number of values of x_j . The ten emails' tables become 3/7, 3/7, 4/7 (spam) and 4/7, 4/7, 3/7 (ham); $(0, 0, 1)$ gets 0.703.

6 Add logs instead of multiplying



Hundreds of factors below 1 underflow: float64 rounds anything under about 5×10^{-324} to 0, and every class ties at 0. A sum of logs keeps the same winner:

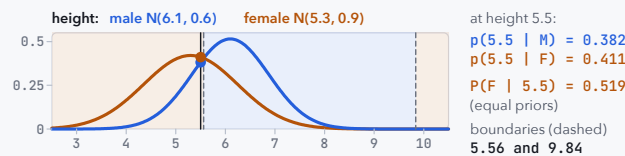
$$\hat{y} = \arg \max_y \left[\log P(y) + \sum_{j=1}^n \log P(x_j \mid y) \right].$$

To get probabilities back, subtract the largest score before exponentiating (log-sum-exp). Smoothing keeps $\log 0$ out.

Real-valued features and graphs of dependence

One Gaussian per class and feature, the discriminative counterpart, and joints factorised along a graph.

7 One normal per class and feature

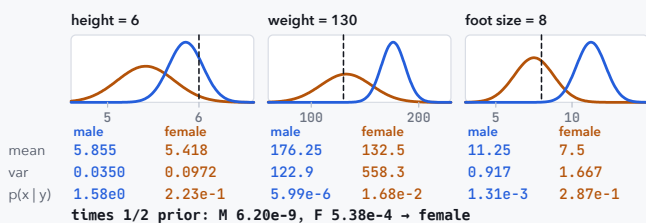


Estimate μ_k and σ_k^2 from the class- k rows and use

$$p(x | C_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} \exp\left(-\frac{(x - \mu_k)^2}{2\sigma_k^2}\right).$$

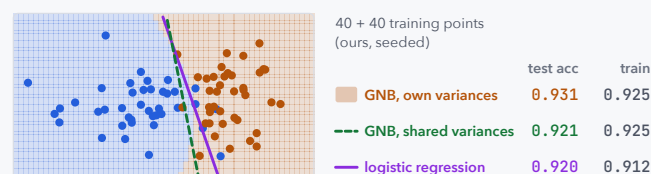
Above, the lecture's heights ($\mathcal{N}(\mu, \sigma^2)$). Unequal variances give a quadratic boundary with two cut points; the wider class wins both tails. Densities can exceed 1: compare them, they are not probabilities.

8 Three features multiply into one score



The lecture's eight people (a widely used public example; feet, pounds, inches), sample variances ($N - 1$). For (6, 130, 8) the female score is 87,000 times the male: $P(F | x) = 0.99999$. Each class is a diagonal-covariance normal.

9 Naive Bayes vs logistic regression



Naive Bayes models $P(x | y)P(y)$; logistic regression fits $P(y | x)$ (the logistic regression sheet). With variances shared by the classes the log-odds is linear, with weights $w_j = (\mu_{1j} - \mu_{0j}) / \sigma_j^2$ set rather than fitted. Naive Bayes needs less data; with more data and correlated features logistic regression usually wins (Ng and Jordan, 2002).

10 A Bayesian network factorises the joint



Nodes are random variables, edges direct influence, no directed cycles:

$$P(X_1, \dots, X_N) = \prod_{k=1}^N P(X_k | \text{parents}(X_k)).$$

Rain or the sprinkler wets the grass, and rain makes the sprinkler unnecessary: $P(S, G, R) = P(G | S, R)P(S | R)P(R)$. Joining every pair saves nothing; a node with k binary parents needs 2^k numbers.

11 Queries sum out what is not observed



The lecture's second network: T and E cause A , E also causes R (say theft, earthquake, alarm, radio report). With R summed out,

$$P(T | A) = \frac{P(A, T)}{P(A)} = \frac{\sum_E P(T, E, A)}{\sum_{T, E} P(T, E, A)}.$$

The slides label this $P(A | T)$, which is $\sum_E P(E)P(A | T, E)$, 0.901 here. The alarm lifts $P(T)$ from 0.01 to 0.365; a report of E cuts it to 0.047: E explains the alarm away.

12 A second test is a second feature



The disease model is the network $D \rightarrow T$. A second test, independent of the first given D , adds a child: $P(D | T_1, T_2) \propto P(D)P(T_1 | D)P(T_2 | D)$. Each positive multiplies the odds by $0.99/0.01 = 99$: from 10^{-4} to 0.0098, then 0.495.

13 Pause and think

Why "naive"? Independent given y .
 "Not bad"? Word order is lost.

Unseen word? Smooth it, or 0 wins.
 NB or logistic? Little data: NB.

Why logs? Products underflow.
 A density above 1? Allowed.

$$\arg \max_y \left[\log P(y) + \sum_j \log P(x_j | y) \right] \cdot \text{count, add 1} \cdot \text{a Gaussian per real feature} \cdot \text{one factor per node}$$

Before: the support vector machines II sheet; next: the time-series forecasting and sequence models sheet. Source: lecture slides "Naive Bayes" and "Bayesian networks", and the Gaussian density from "Multivariate normal distribution" (nipunbatra.github.io/ml-teaching). The fourth word, the per-word probabilities, the 2D data and the network's tables are ours (illustrative, seeded). Corrected: the spam estimates and scores, the female rows of the eight-person table, the male foot-size variance (0.917, not 0.97), $P(A | T)$ written for $P(T | A)$, and the test weight (130 lb, not 13.5).